

Course Introduction

Genya Ishigaki

[Having a challenging time to pronounce my name?](#)

San José State University

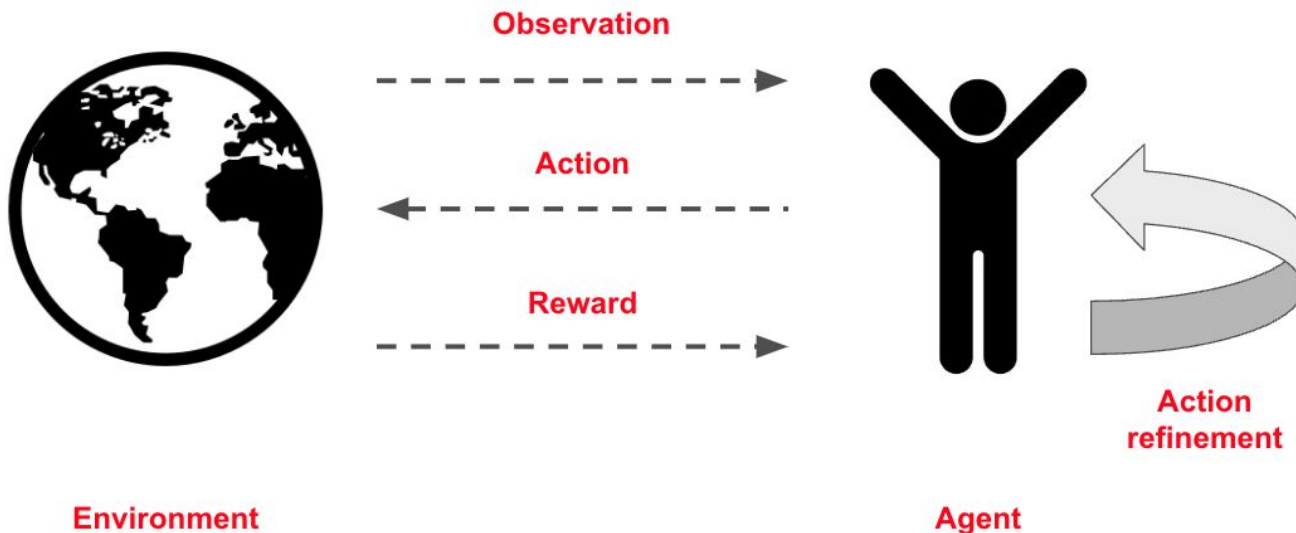
CS272.01 Reinforcement Learning and Sequential Decision Making



Introduction to reinforcement learning, deep reinforcement learning, other online learning algorithms, and their applications.

This course is...

- CS 272 Reinforcement Learning and Sequential Decision Making
- RL is an area of machine learning
- An agent learns a sequence of optimal actions to realize a goal



Today's Agenda

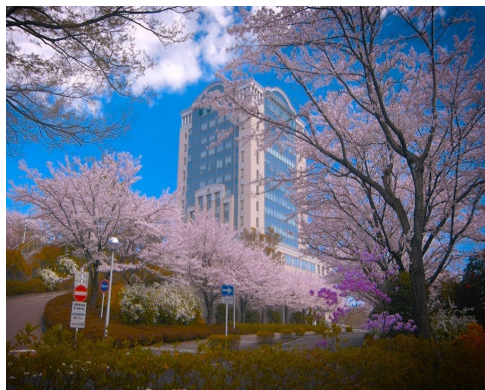


- My Introduction
 - Breakout: Getting to know new fellow students
- My research areas
- Course syllabus
- Announcements
- Q&A

About Me



- Born and raised in Japan
- Obsessed with LEGO during the childhood
- Majoring CS from my bachelors
 - M.S. & B.S. in Engineering from Soka University
- Came to US for my PhD in 2016
 - Ph.D. & M.S. in CS from University of Texas at Dallas
- Joined SJSU in August, 2021



Breakout: Getting to know new fellow students



- Form a group (with 4 or 5 ppl around you)
- Introduce yourself!

About Me



About Me: Research Interest

- 5G and Beyond Networking & Network Slicing
- Deep Reinforcement Learning

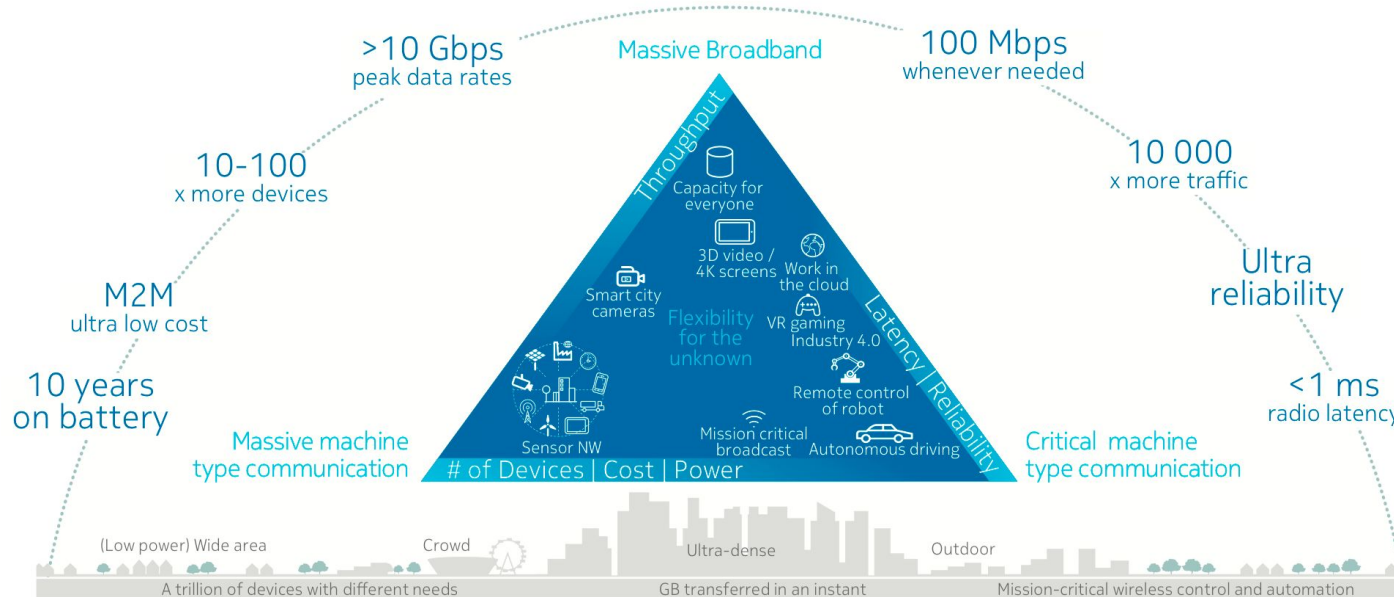


Figure 1. 5G will enable very diverse use cases with extreme range of requirements

About Me: Research Interest

- 5G and Beyond Networking & Network Slicing
- Deep Reinforcement Learning



Documentary: "AlphaGo - The Movie", <https://www.youtube.com/watch?v=WXuK6gekU1Y>

For more information



sjsu interconnect



All



News



Images



Maps



Shopping



More

Tools

About 119,000 results (0.52 seconds)

<https://sjsu-interconnect.github.io> :

interconnect lab | SJSU interconnect lab researches next ...

SJSU interconnect lab researches next-generation telecommunication networks, with an emphasis on developing combinatorial optimization and machine learning ...

Course Syllabus

Check Canvas!

Wednesday Quizzes



- 10 quizzes (Please check the syllabus for the schedule)
- Basic questions about the concepts and coding
- A very small impact on the overall grade
- A review session right after the quiz during the class
 - Questions/Answers will NOT be published on Canvas
 - Please take notes during the review
- An opportunity to check your understanding and ask questions

Programming Assignment Format



- Base codes will be provided at each PA class
- We will do coding together in class
 - Ready to bring your laptop with
 - A Python environment (See the next slide)
 - Zoom
 - Laptop lending at [Student Computing Services](#)
- You need to complete the rest after the class by yourself
- A submission will be evaluated based on python unit testing
 - Actual test cases will not be revealed
 - A submission needs to follow a specified format for the auto evaluation
 - If your code cannot be run due to the format issue, the submission may not be correctly evaluated
 - Also, the quality of the codes will be evaluated

- OOP in Python: <https://realpython.com/python3-object-oriented-programming/>
- Virtual python environment (make sure you have Python 3.x)
 - venv: <https://docs.python.org/3/library/venv.html>
 - Anaconda: <https://www.anaconda.com/>
 - mini-forge (especially for M1 Mac):
<https://github.com/conda-forge/miniforge>
- IDE
 - PyCharm: <https://www.jetbrains.com/pycharm/>
 - VSCode: <https://code.visualstudio.com/docs/languages/python>
- My environment: VSCode + Python 3.10.5 (through mini-forge)

Programming Assignment Warnings



- Code similarity will be checked using a plagiarism software
 - If the score exceeds 20%, you will receive an F grade for the assignment
- You can discuss the problem with other fellow students
- Never show/share your codes
- Do not write codes (or maybe pseudo codes) with fellow students

Exam Format



- Canvas Online Exam in Class
- Conceptual questions
- Simple calculation/algorithm execution questions
- Scratch paper section for partial credits
- No cumulative final

> Question 12 0 pts

Scratch Paper (Optional):

Edit View Insert Format Tools Table

12pt ▾ Paragraph ▾ | **B** *I* U A ▾ ▾ T² ▾ | ⋮

p 0 words ⋮

Course Announcements



Canvas Message

- Please expect a response within a few days
- Please do not send an email for course-related questions

No Email.

Any Questions?